AP PRECALCULUS

Unit 1 • Worksheet 1.4A • Answer key

Worksheet 1.4A • Polynomial Functions and Rates of Change

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What is the greatest possible number of local extrema of a polynomial of degree 5?

(A) 5

(B) 4

(C) 3

(D) 6

Solution

Answer: (B). At most $n - 1$.

Question 2

NO CALCULATOR

What is the degree and leading coefficient of $p(x) = -3x(x+2)^2(1-x)$?

(A) degree 4, leading coefficient -3

(B) degree 4, leading coefficient 3

(C) degree 3, leading coefficient 3

(D) degree 5, leading coefficient -3

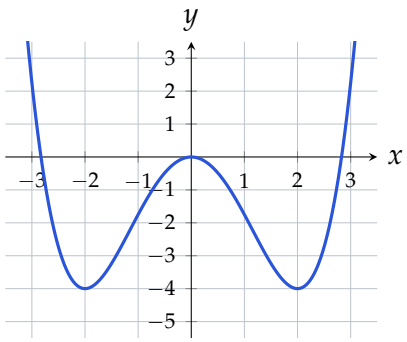
Solution

Answer: (B). Leading term $-3x \cdot x^2 \cdot (-x) = 3x^4$.

Question 3

NO CALCULATOR

The graph of $p(x) = \frac{1}{4}x^4 - 2x^2$ is shown. What is the absolute minimum value of p?



(A) 0
(B) -2
(C) -5
(D) -4

Solution
Answer: (D). Local minima $(\pm 2, -4)$ are the lowest points.

Question 4

NO CALCULATOR

Using the same graph, what is the absolute maximum of p on $[-1, 3]$?

(A) 0
(B) 2.25
(C) -1.75
(D) 4

Solution
Answer: (B). Compare $p(-1) = -1.75$, $p(0) = 0$, $p(2) = -4$, $p(3) = 20.25 - 18 = 2.25$.

Question 5 (Free response)

NO CALCULATOR

The polynomial $p(x) = x^3 - 6x^2 + 9x$ has a local maximum at $(1, 4)$, a local minimum at $(3, 0)$, and a point of inflection at $(2, 2)$.

(a) On which intervals is p increasing? Decreasing?
(b) Find the absolute maximum and minimum values of p on $[0, 4]$.
(c) On which interval is p decreasing and concave up? Describe how p changes there.

Solution
(a) Increasing on $(-\infty, 1)$ and $(3, \infty)$; decreasing on $(1, 3)$.
(b) $p(0) = 0$, $p(1) = 4$, $p(3) = 0$, $p(4) = 4$: maximum 4, minimum 0.
(c) $(2, 3)$: p decreases more and more slowly.

Question 6 (Free response) NO CALCULATOR

Sketch the graph of a polynomial of degree 4 with a positive leading coefficient, local minima at $(-2, -3)$ and $(2, -3)$, and a local maximum at $(0, 1)$. How many points of inflection must it have?

Solution

A "W"-shaped graph through those three turning points, rising at both ends. It must have at least two points of inflection: one between $x = -2$ and $x = 0$, one between $x = 0$ and $x = 2$.